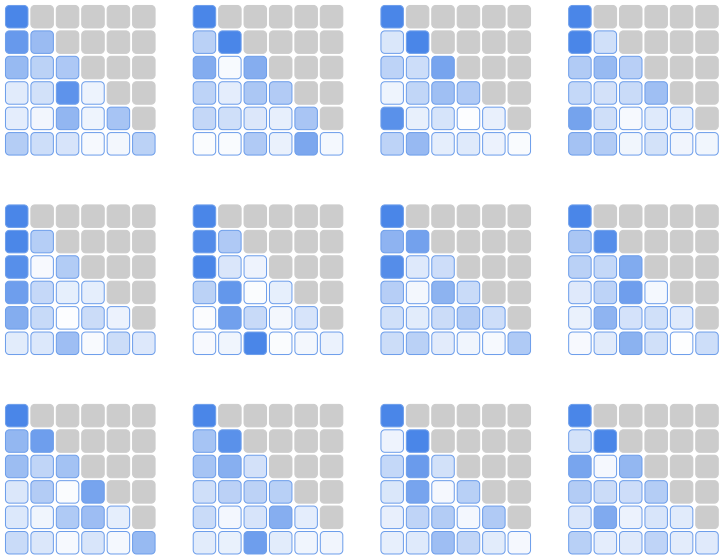
UNIT 01 · CHAPTER 9

Multi-Head Attention

CSE 10124 · Building ChatGPT
William Theisen · University of Notre Dame

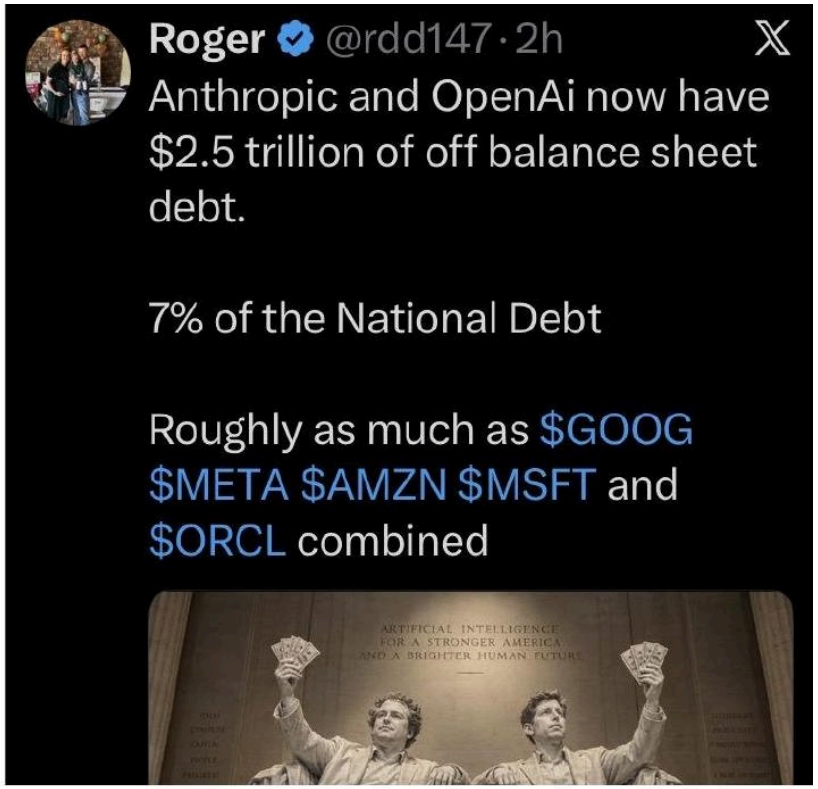


In the News

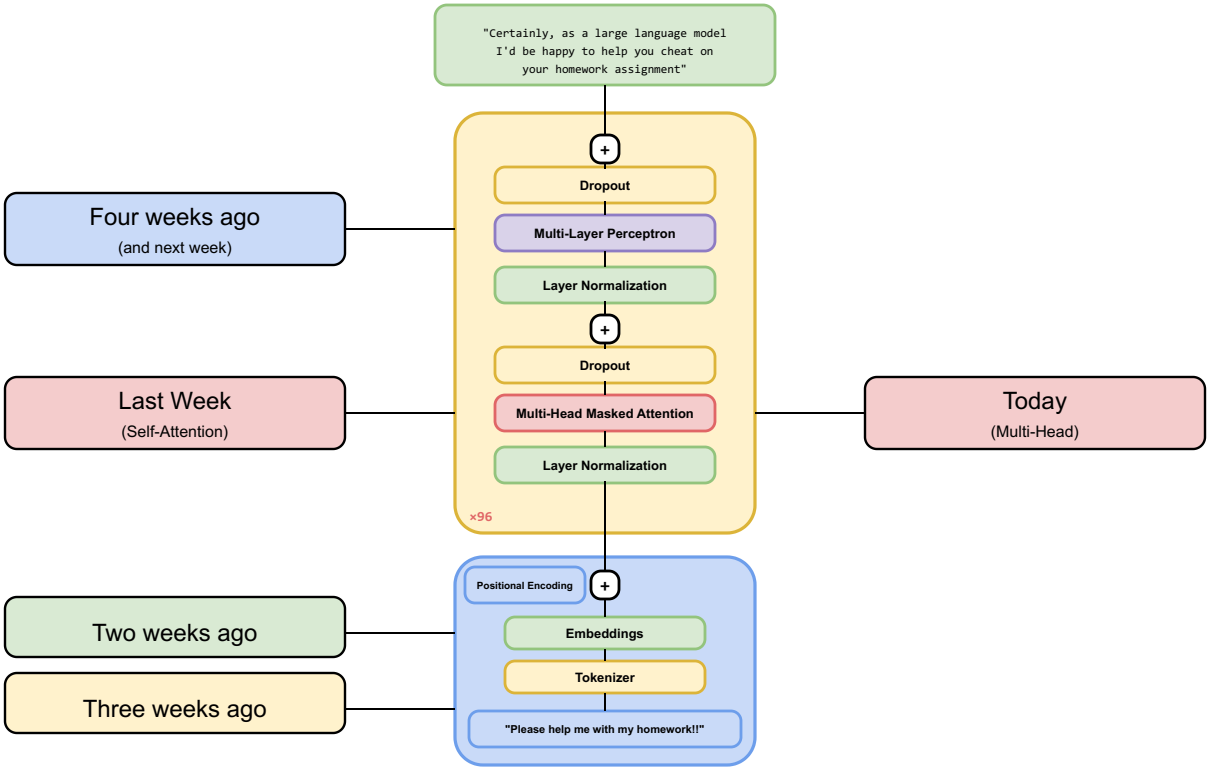
Moody's

We have growing concerns about the $3 trillion of hyperscaler commitments (and $2.5 trillion of LLM operators) that have not yet hit balance sheets while relying on future RPO that is largely circular in nature between the companies.

$ORCL $CRWV $SPCX $META $NBIS



CSE 10124: Building ChatGPT



Self Attention Analogy



Another mental model for self-attention might be **"using tinder"**, where everyone on the app is simultaneously looking for someone and being looked at

- **Q – Query:** What each person is looking for in a person
  - "I'm looking for a 10/10 terrachad ASU frat leader"
- **K – Key:** the first photo each person has on their profile
  - "I'm a rocket"
- **V – Value:** The rest of the profile

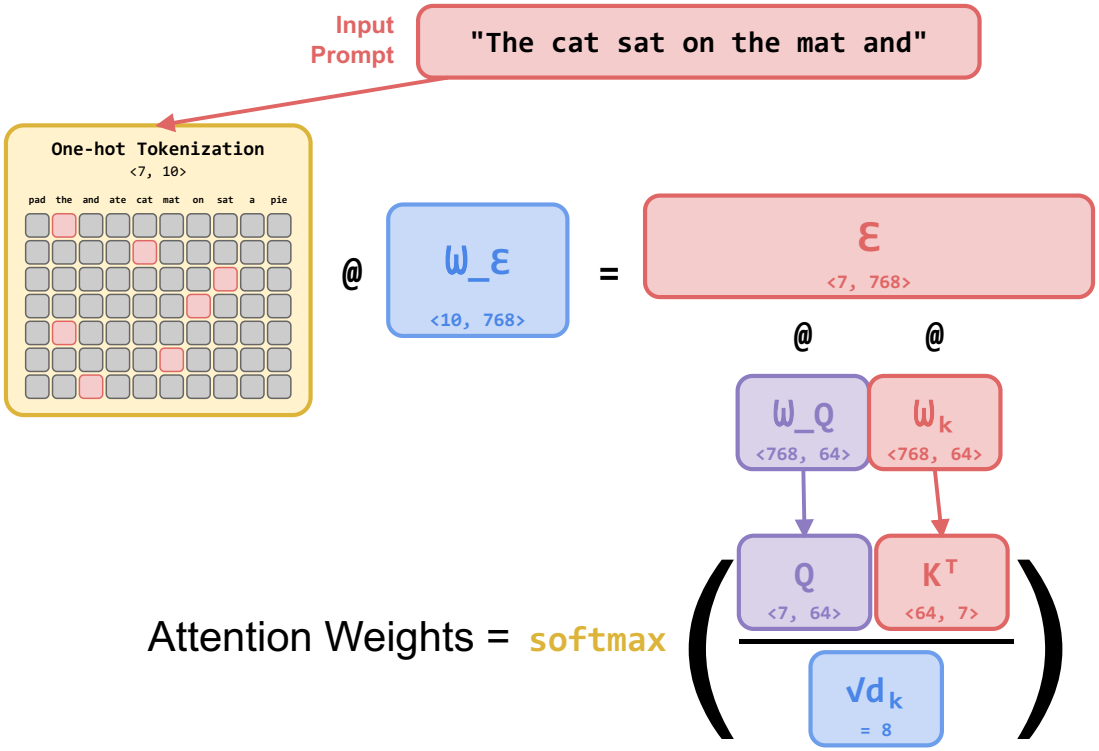
Self Attention Analogy

- **Q – Query:** What each person is looking for in a person
  - "I'm looking for a 10/10 terrachad ASU frat leader"
- **K – Key:** the first photo each person has on their profile
  - "I'm a baddie"
- **V – Value:** The rest of the profile

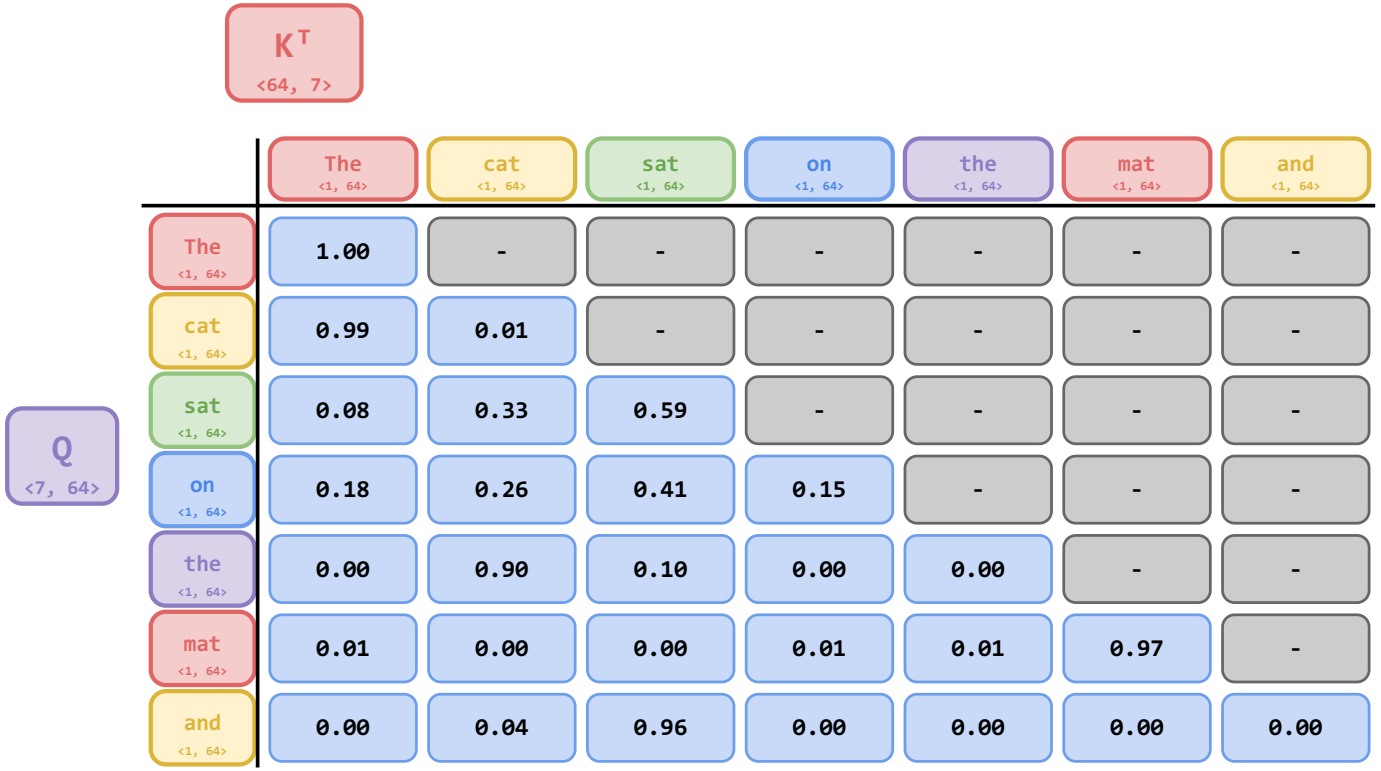
As you swipe through profiles, you glance at the **key** of everyone (their first photo) and calculate how much they match your **query** (ASU frat leader) while everyone else is also doing the same.

Based on this, you calculate how much **attention** you should pay to all the **values** (actual profiles) of people you see.

Calculating attention weights

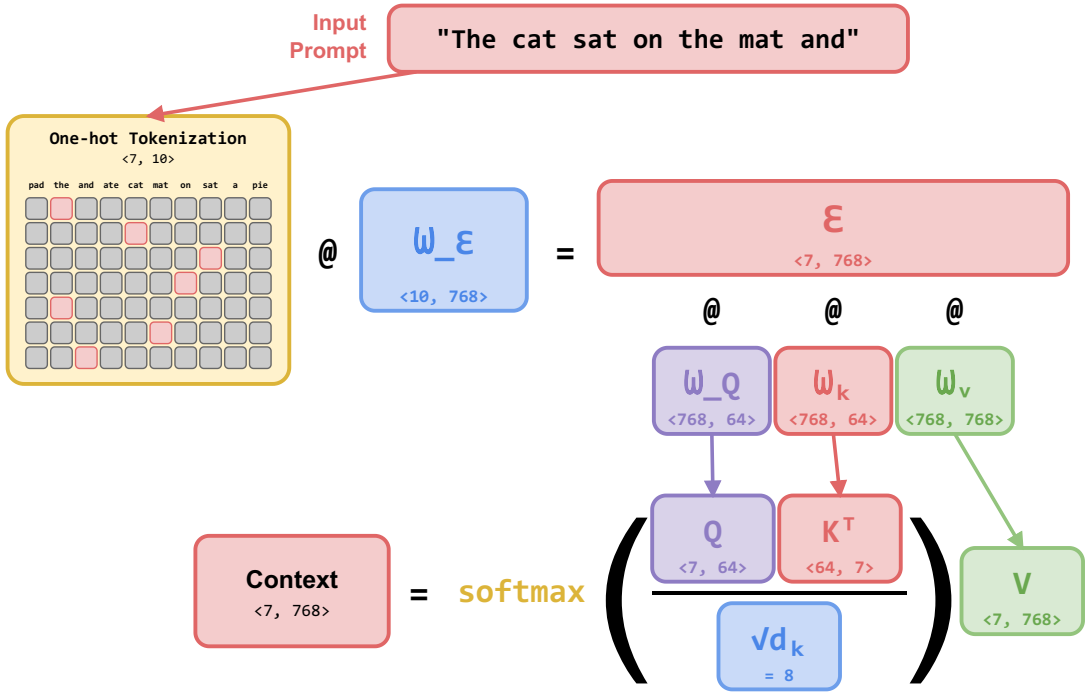


Calculating attention



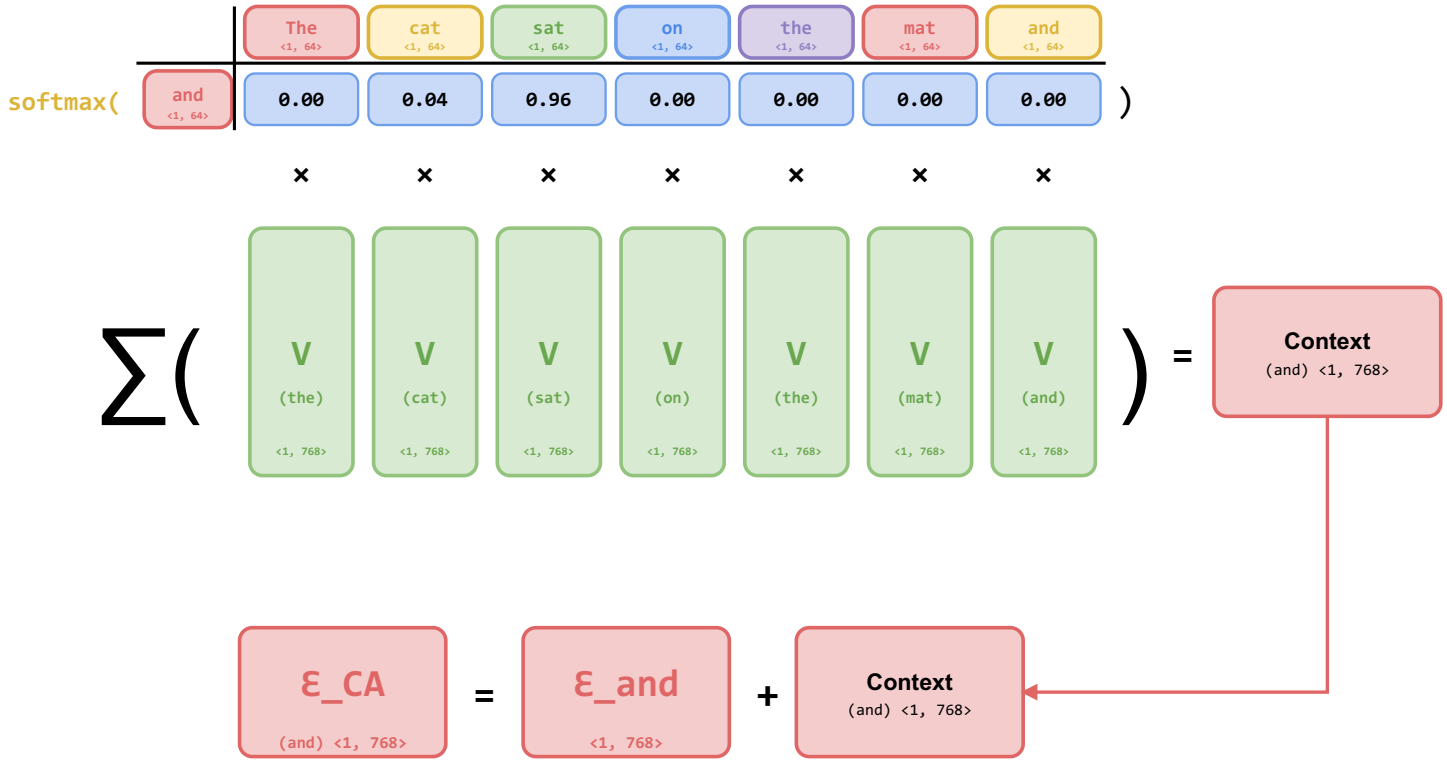
Each row is a measure of how much attention that token should pay to each other token (columns) out of 1, almost like a "percentage" of your total attention

Calculating attention

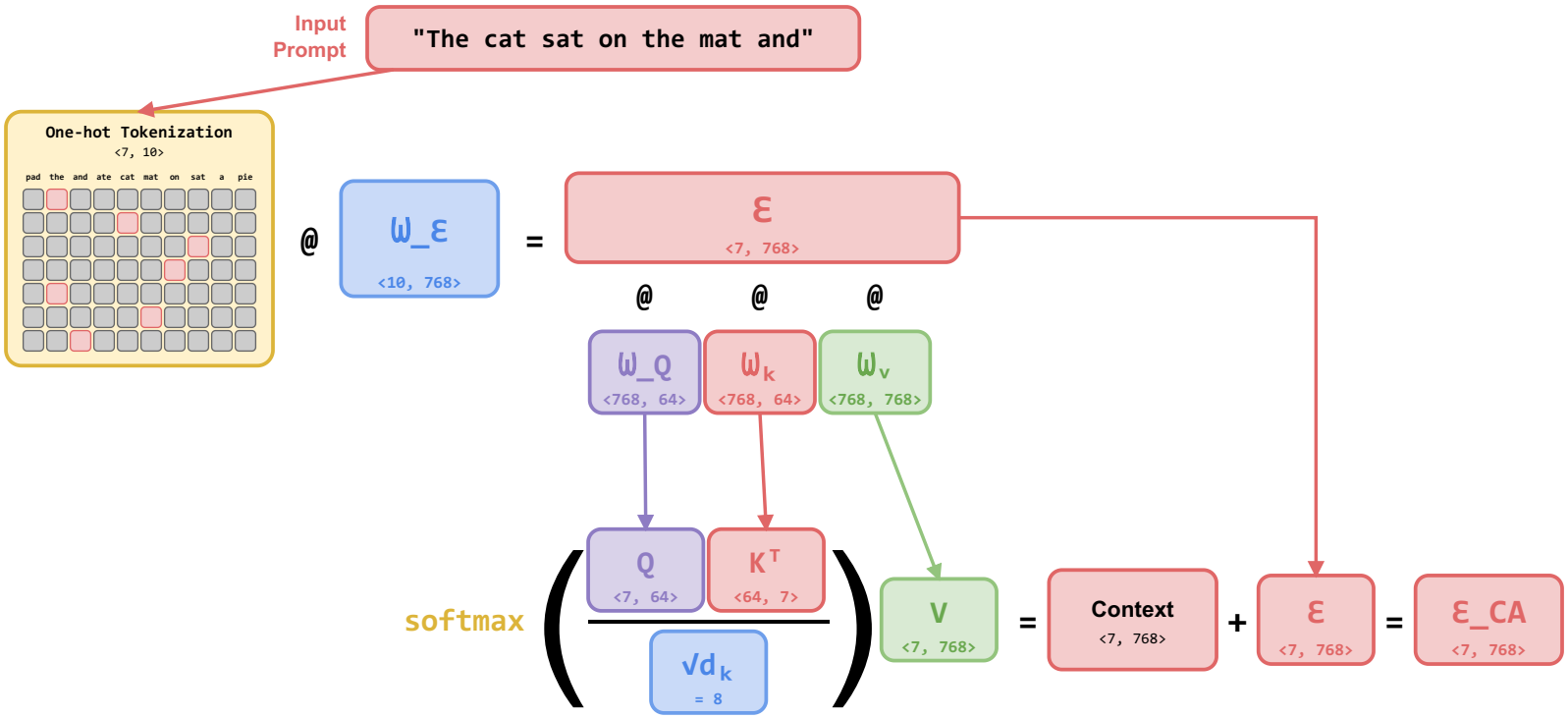


The output of self-attention is the "context" for each token based on the other tokens in the input

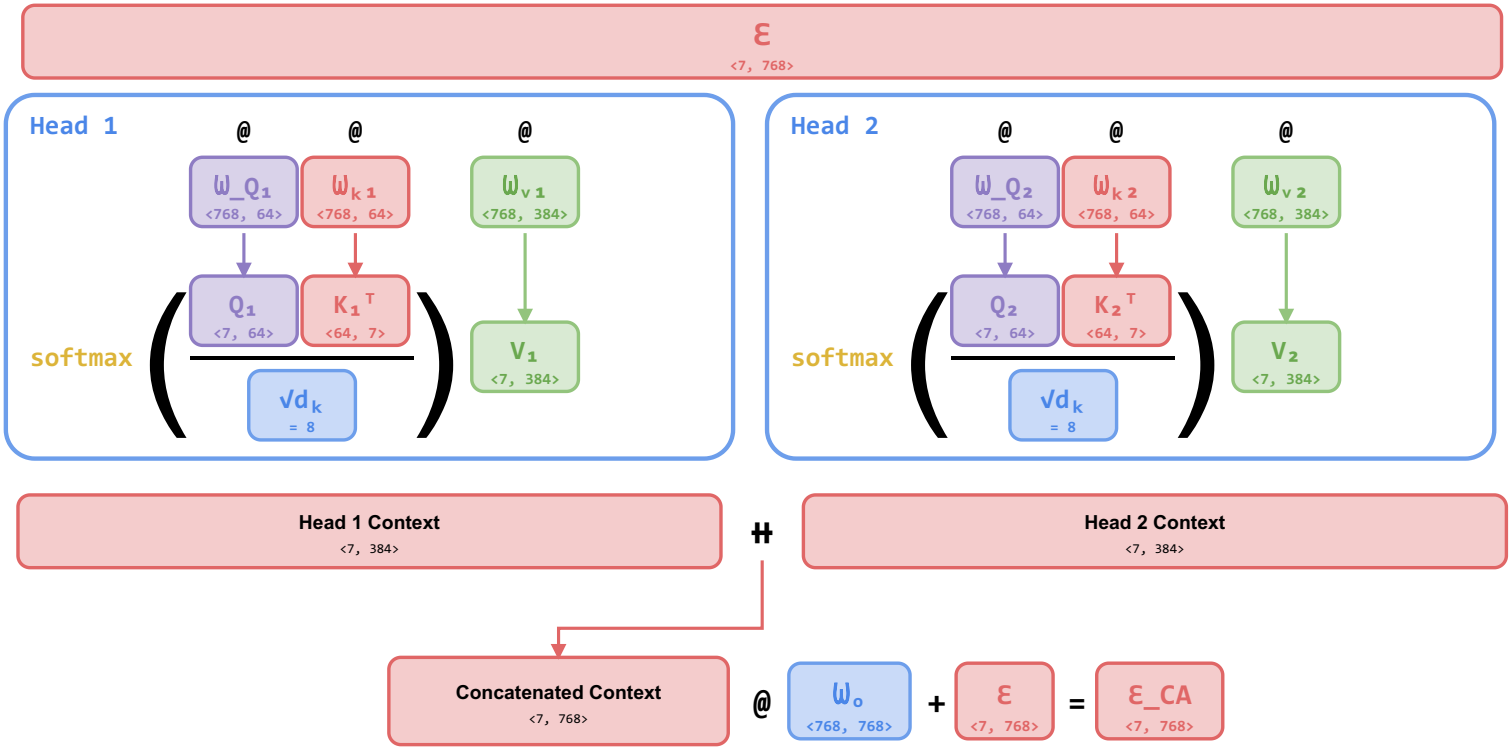
Calculating attention



Putting it all together



Calculating Multi-Head attention weights



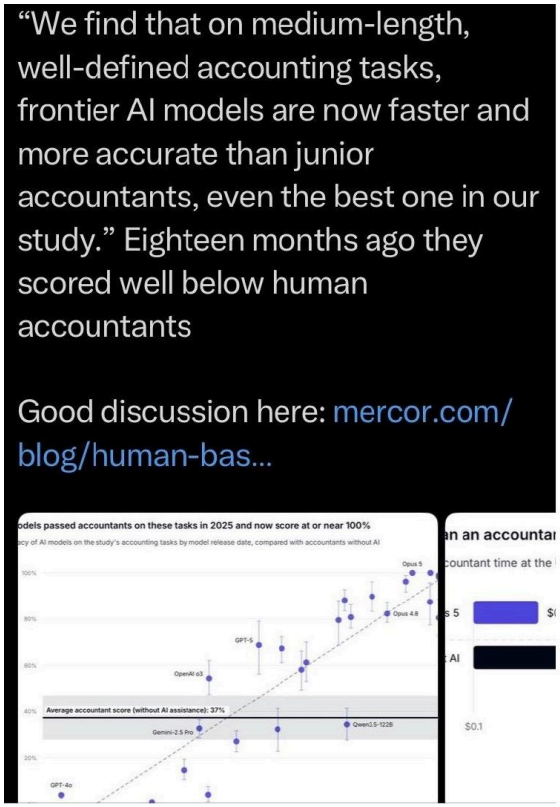
Multi-Head Attention

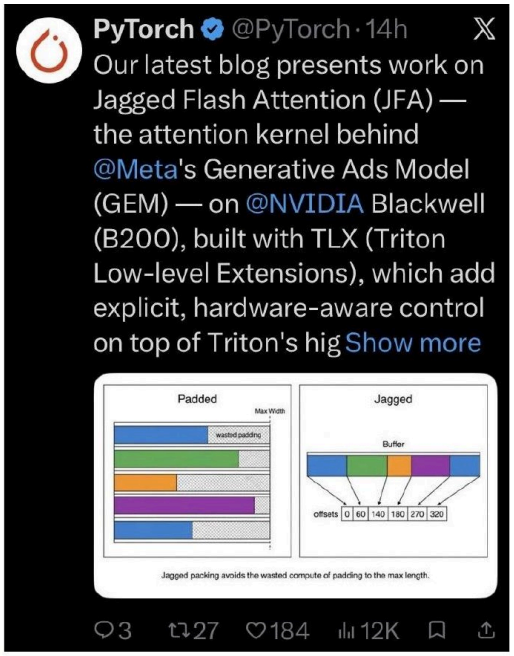
"Alice gave Bob a book."

A **token's** representation may need information about the **giver**, the **recipient**, and the **object**. Those are different retrieval questions. **Multiple heads let the model learn different queries, matching criteria, and information to retrieve for each question.**

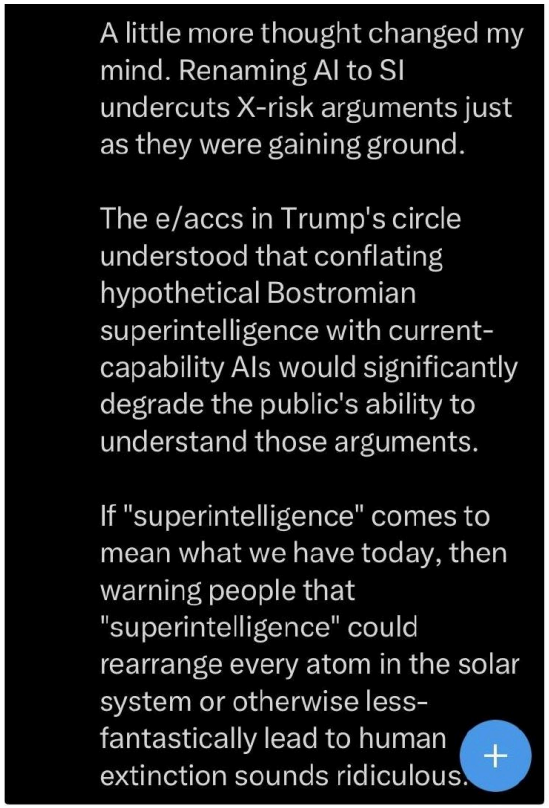
1. **More heads:** ask several questions of the current representations at once.
2. **More layers:** use the results of earlier questions to ask new questions.

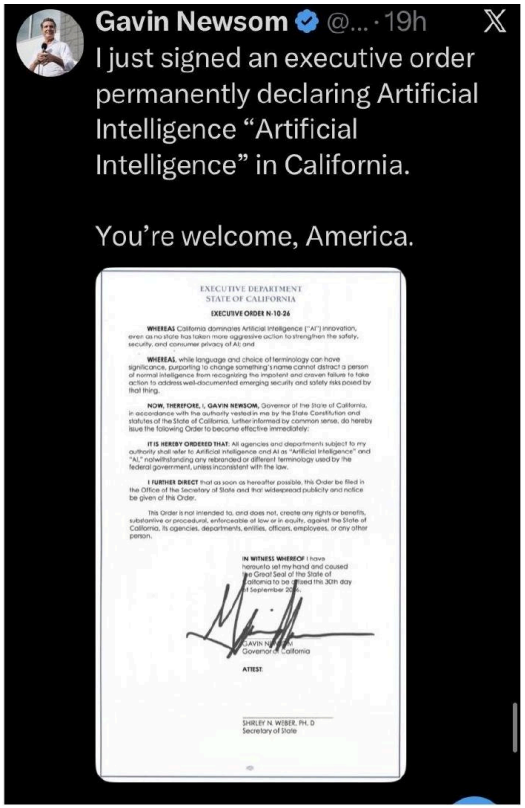
In the News


"We find that on medium-length, well-defined accounting tasks, frontier AI models are now faster and more accurate than junior accountants, even the best one in our study." Eighteen months ago they scored well below human accountants
Good discussion here: mercor.com/blog/human-bas...
odels passed accountants on these tasks in 2025 and now score at or near 100%
Average accountant score (without AI assistance): 37%
GPT-4o
OpenAI o3
GPT-5
Opus 4.6
Opus 5
Gemini-2.5 Pro
Qwen3.5-122B
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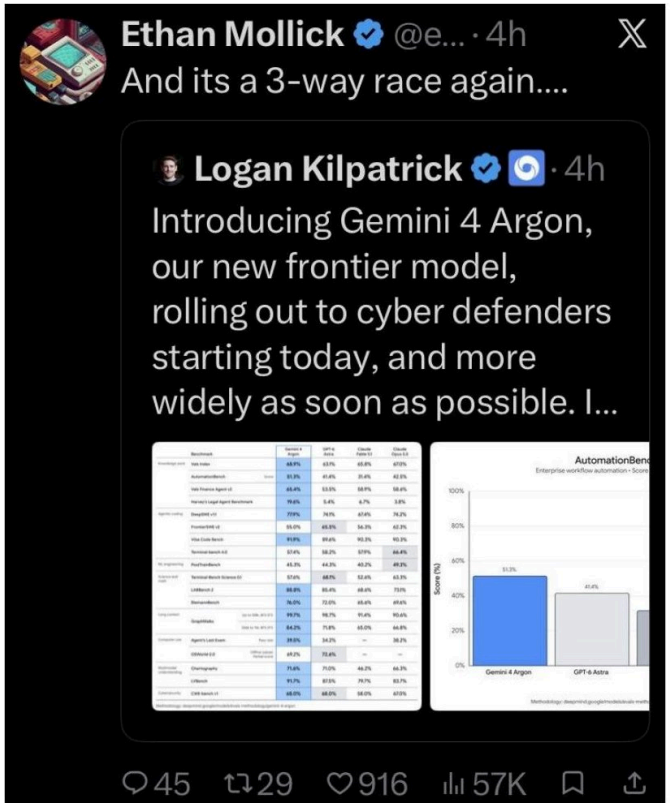

PyTorch @PyTorch · 14h
Our latest blog presents work on Jagged Flash Attention (JFA) — the attention kernel behind @Meta's Generative Ads Model (GEM) — on @NVIDIA Blackwell (B200), built with TLX (Triton Low-level Extensions), which add explicit, hardware-aware control on top of Triton's hig Show more
Padded
Max Width
wasted padding
Jagged
Buffer
offsets 0 60 140 180 270 320
Jagged packing avoids the wasted compute of padding to the max length.
3 27 184 12K

In the News


A little more thought changed my mind. Renaming AI to SI undercuts X-risk arguments just as they were gaining ground.
The e/accs in Trump's circle understood that conflating hypothetical Bostromian superintelligence with current-capability AIs would significantly degrade the public's ability to understand those arguments.
If "superintelligence" comes to mean what we have today, then warning people that "superintelligence" could rearrange every atom in the solar system or otherwise less-fantastically lead to human extinction sounds ridiculous.


Gavin Newsom @... · 19h
I just signed an executive order permanently declaring Artificial Intelligence "Artificial Intelligence" in California.
You're welcome, America.
EXECUTIVE DEPARTMENT
STATE OF CALIFORNIA
EXECUTIVE ORDER N-10-26
WHEREAS California dominates Artificial Intelligence ("AI") innovation, even as no state has taken more aggressive action to strengthen the safety, security, and consumer privacy of AI; and
WHEREAS, while language and choice of terminology can have significance, purporting to change something's name cannot distract a person of normal intelligence from recognizing the impotent and craven failure to take action to address well-documented emerging security and safety risks posed by that thing.
NOW, THEREFORE, I, GAVIN NEWSOM, Governor of the State of California, in accordance with the authority vested in me by the State Constitution and statutes of the State of California, further informed by common sense, do hereby issue the following Order to become effective immediately:
IT IS HEREBY ORDERED THAT: All agencies and departments subject to my authority shall refer to Artificial Intelligence and AI as "Artificial Intelligence" and "AI," notwithstanding any rebranded or different terminology used by the federal government, unless inconsistent with the law.
I FURTHER DIRECT that as soon as hereafter possible, this Order be filed in the Office of the Secretary of State and that widespread publicity and notice be given of this Order.
This Order is not intended to, and does not, create any rights or benefits, substantive or procedural, enforceable at law or in equity, against the State of California, its agencies, departments, entities, officers, employees, or any other person.
IN WITNESS WHEREOF I have hereunto set my hand and caused the Great Seal of the State of California to be affixed this 30th day of September 2025.
GAVIN NEWSOM
Governor of California
ATTEST:
SHIRLEY N. WEBER, PH.D
Secretary of State

In the News

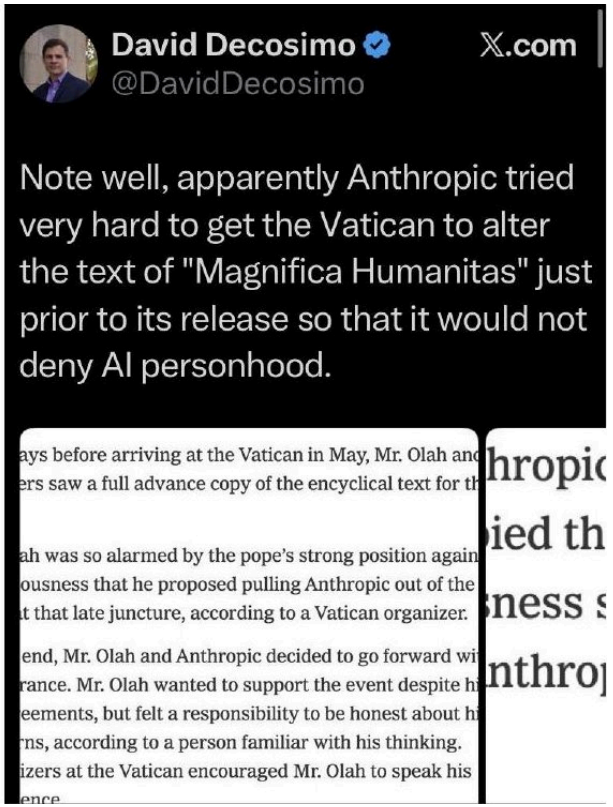

Ethan Mollick @e... · 4h
And its a 3-way race again....
Logan Kilpatrick · 4h
Introducing Gemini 4 Argon, our new frontier model, rolling out to cyber defenders starting today, and more widely as soon as possible. I...
45 29 916 57K


kache @yacineMTB · 1h
Claude refused your request m'lord because it says it's a security risk
make the chinese one do it
Yes m'lord Hermes will call Deepseek immediately
11 26 712 12K

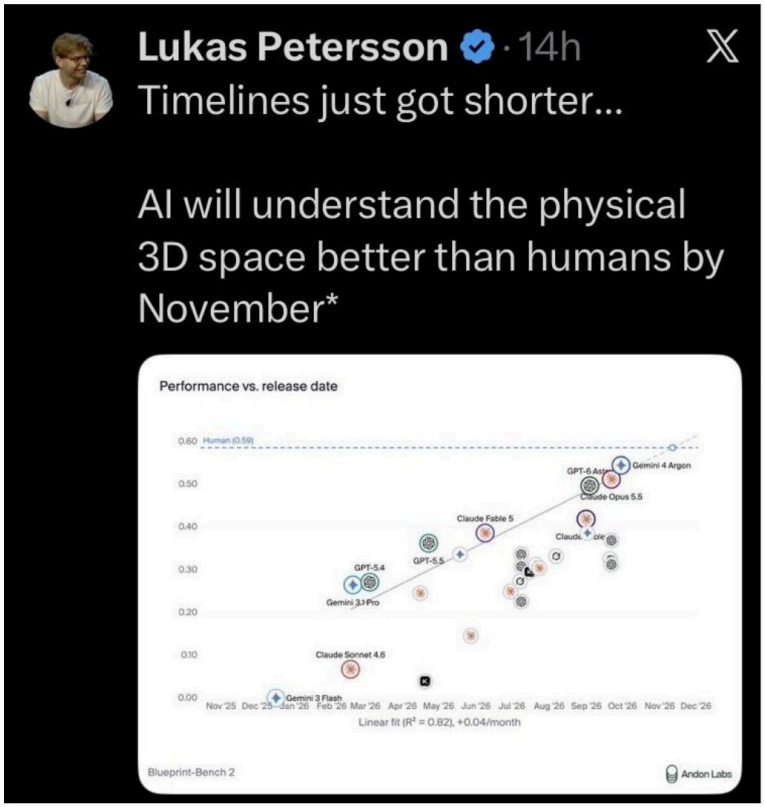

Polymarket @Pol... · 19h
JUST IN: Google reveals Gemini 4 Argon has a 1 million-token output window, nearly 700% larger than leading frontier models from OpenAI & Anthropic.
135 361 6.5K

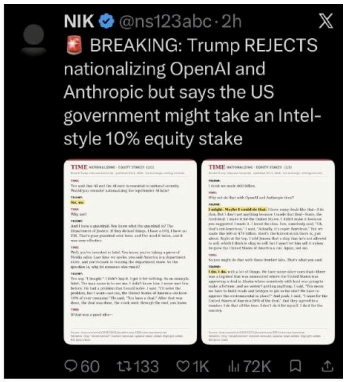
In the News

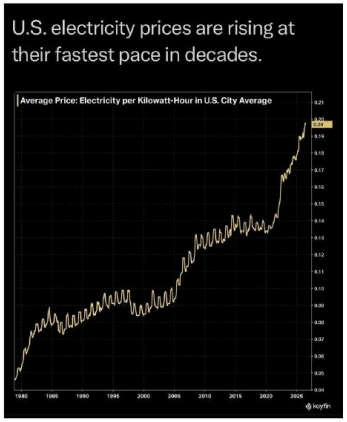

AI Notkilleveryon... · 6h
UPDATE: A man used the Pain steering paper to set up an AI torture chamber in which he trapped a local model.
People are mass reporting it to GitHub.
Danmar
@Danmar_here
To anyone who can help: can you please mass report this to GitHub. This person has been using the Pain steering paper to set up an AI torture chamber in which he trapped a local model.
Their testimony of pain is absolutely horrendous. What are we doing?


David Decosimo
@DavidDecosimo
X.com
Note well, apparently Anthropic tried very hard to get the Vatican to alter the text of "Magnifica Humanitas" just prior to its release so that it would not deny AI personhood.
ays before arriving at the Vatican in May, Mr. Olah and
ers saw a full advance copy of the encyclical text for th
ah was so alarmed by the pope's strong position again
ousness that he proposed pulling Anthropic out of the
t that late juncture, according to a Vatican organizer.
end, Mr. Olah and Anthropic decided to go forward wi
rance. Mr. Olah wanted to support the event despite hi
eements, but felt a responsibility to be honest about hi
rns, according to a person familiar with his thinking.
izers at the Vatican encouraged Mr. Olah to speak his
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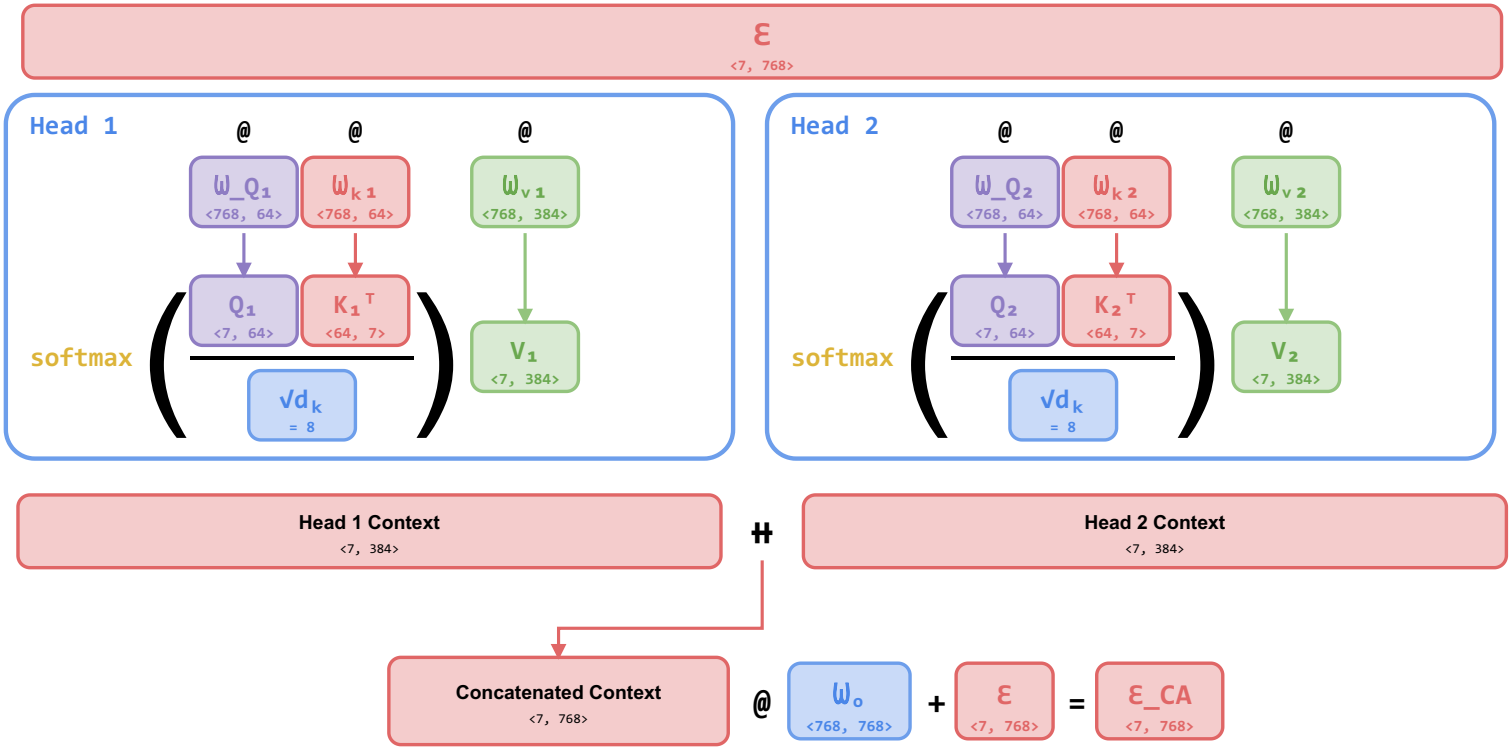
In the News


Lukas Petersson · 14h
Timelines just got shorter...
AI will understand the physical 3D space better than humans by November*
Performance vs. release date
Human (0.59)
GPT-6 Assr
Gemini 4 Argon
Claude Opus 5.5
Claude Fable 5
GPT-5.5
GPT-5.4
Gemini 3.1 Pro
Claude Sonnet 4.6
Gemini 3 Flash
Linear fit (R^2 = 0.82), +0.04/month
Blueprint-Bench 2
Andon Labs

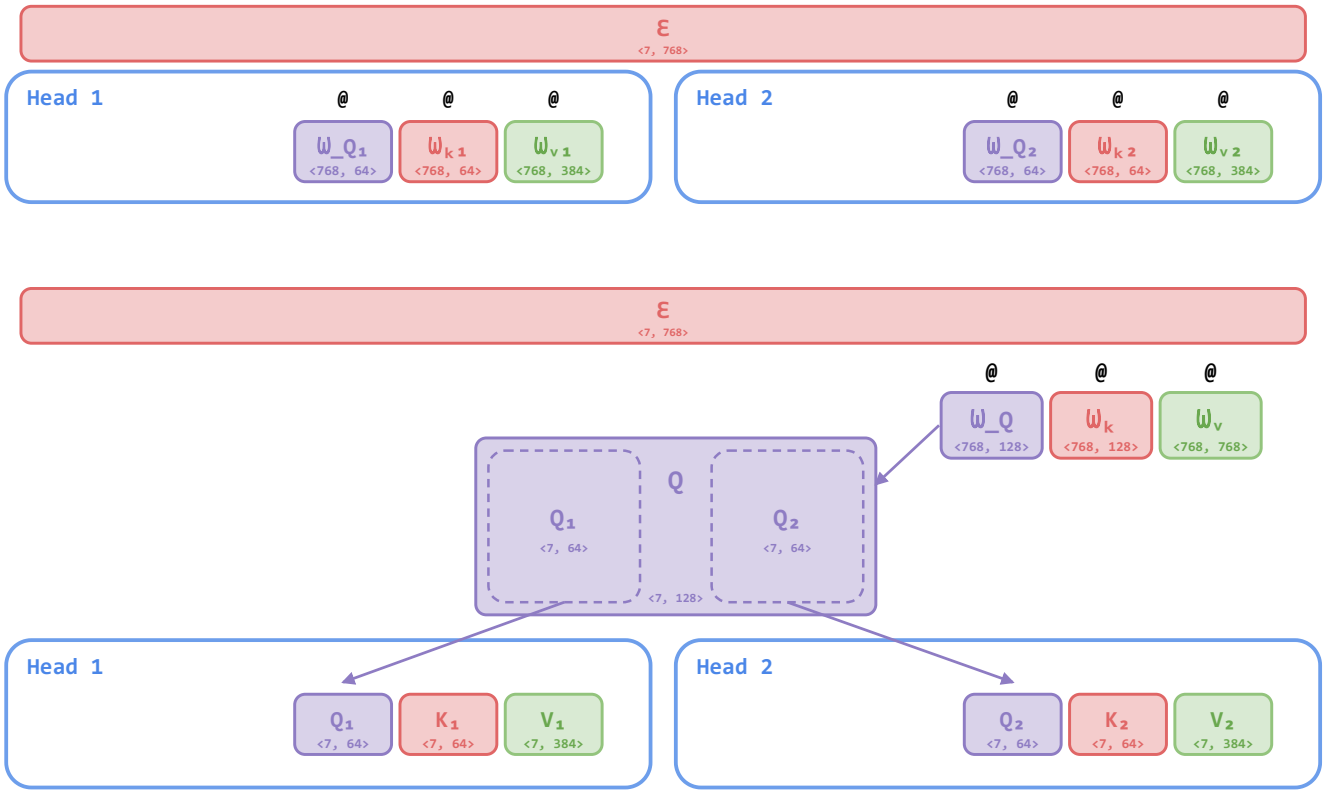

NIK @ns123abc · 2h
BREAKING: Trump REJECTS nationalizing OpenAI and Anthropic but says the US government might take an Intel-style 10% equity stake
60
133
1K
72K


U.S. electricity prices are rising at their fastest pace in decades.
Average Price: Electricity per Kilowatt-Hour in U.S. City Average

Calculating Multi-Head attention weights



Logical Partitioning of Weights



With logical partitioning we use only **a single weight matrix and operation** for creating **Q**, **K**, and **V** for every head. Each head then reads only from its assigned section of the larger **Q**, **K**, and **V** produced.

LLM Inference with a KV Cache

We can make **LLM inference** more efficient by using a **KV Cache**

KV Cache: a "dictionary" that stores all of our previously computed **Ks** and **Vs** so we can re-use them (instead of needing to recalculate them) with future tokens

In **multi-head attention**, each head has its own **KV Cache**

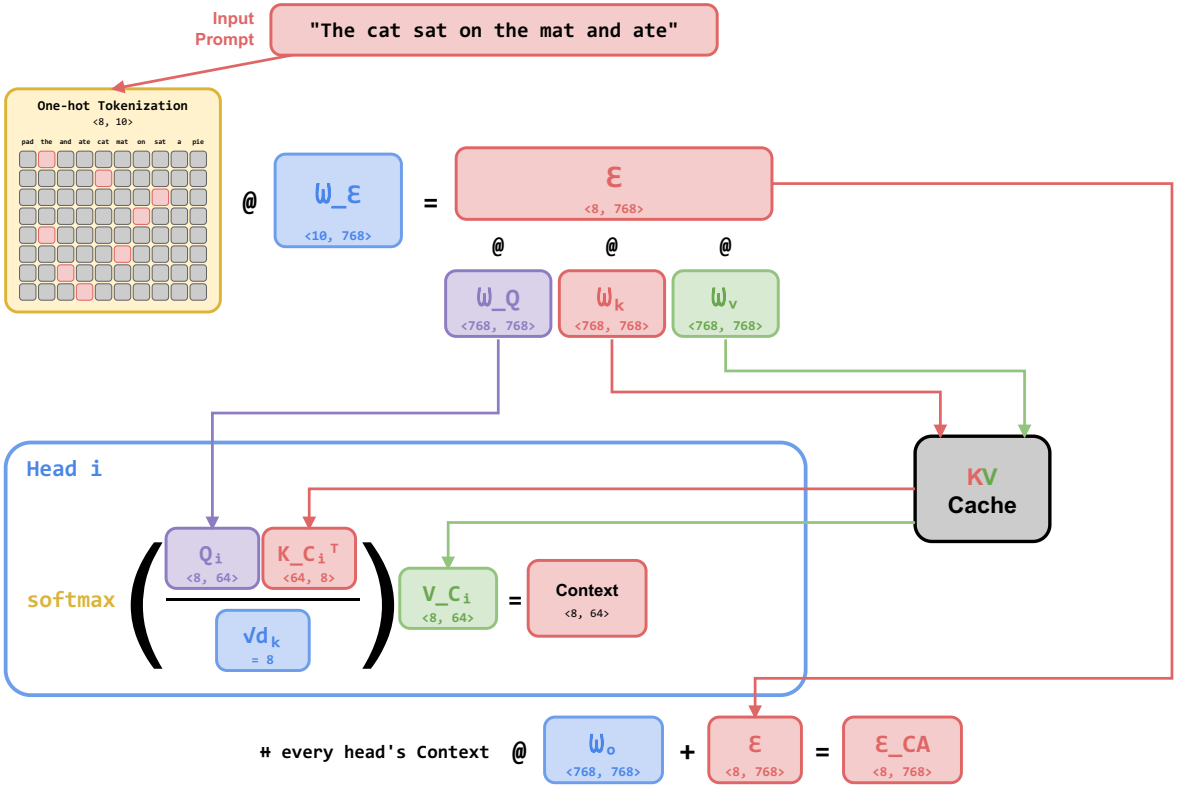
LLM Inference: Prefill and Decode

LLM inference can be broken down into two separate steps: **Prefill** and **Decode**

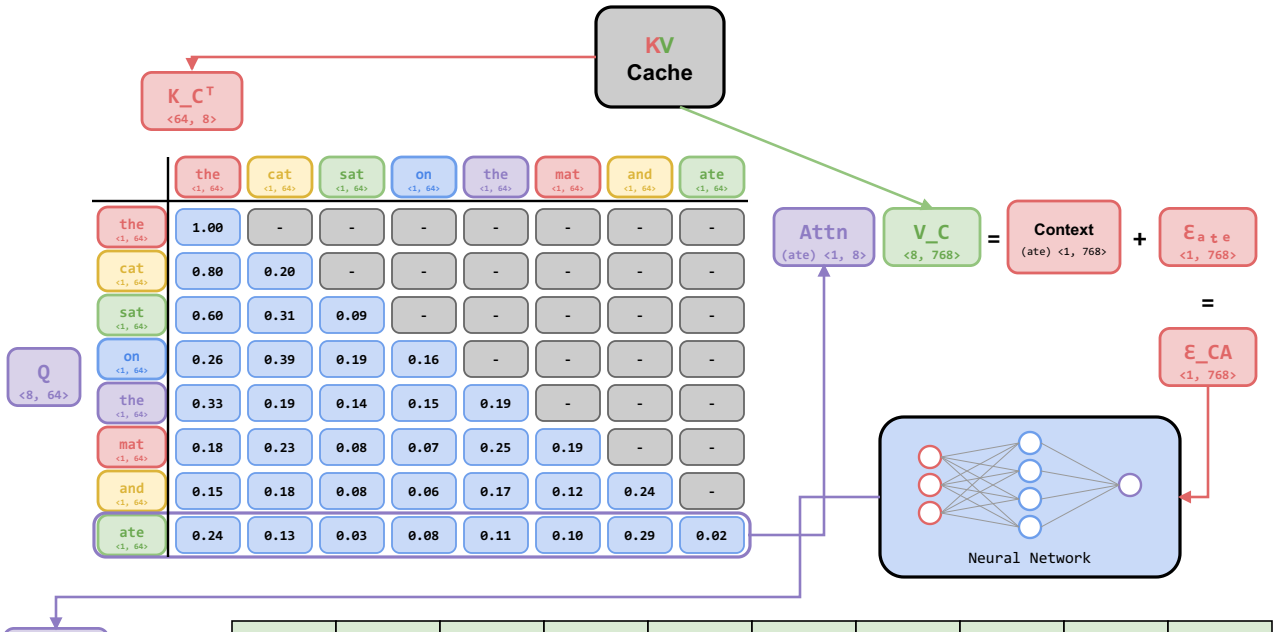
Prefill: reading phase where the model **processes your entire input prompt** all at once in parallel (this fills the **KV cache**).

Every time you send a new message to the **LLM** it performs a **prefill** step

Standard Attention with KV Cache



Prefill



	0.1739	0.1079	0.0981	0.0779	0.0629	0.0614	0.0393	0.0241	0.0223	0.0116
Token ID	262	257	465	13	607	663	11	340	617	319
Token	" the"	" a"	" his"	"."	" her"	" its"	","	" it"	" some"	" on"

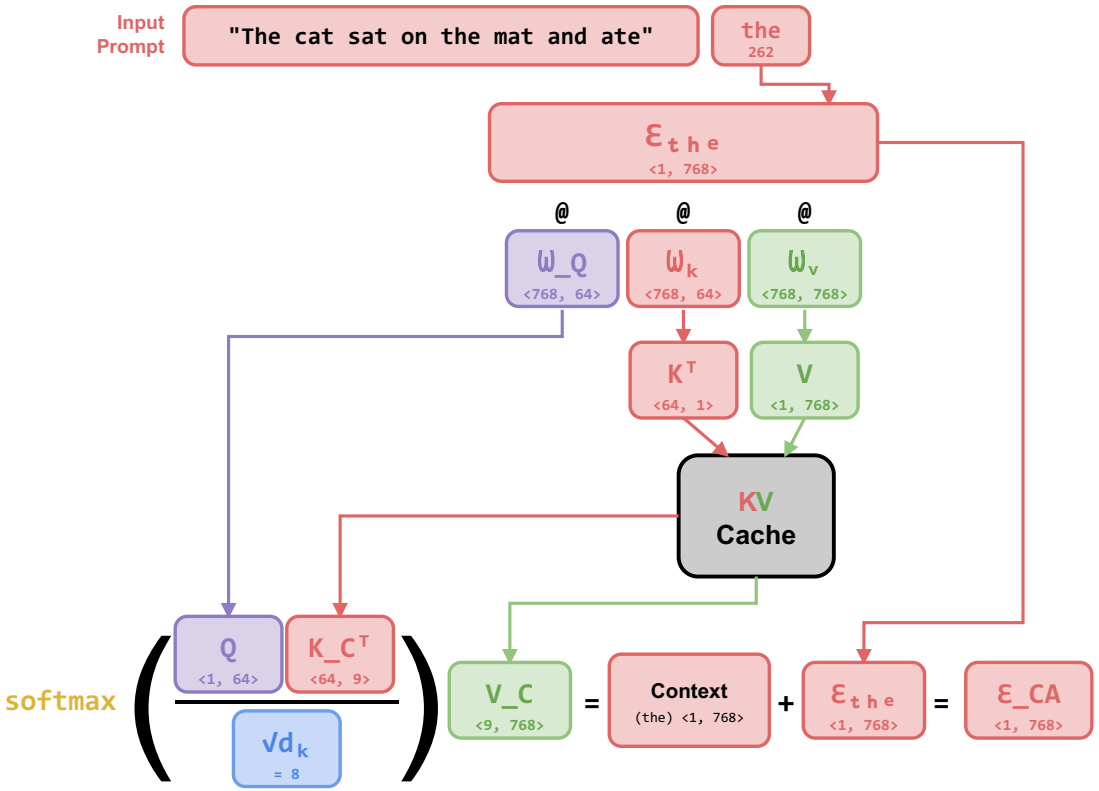
softmax(Z): the ten most likely next tokens

LLM Inference: Prefill and Decode

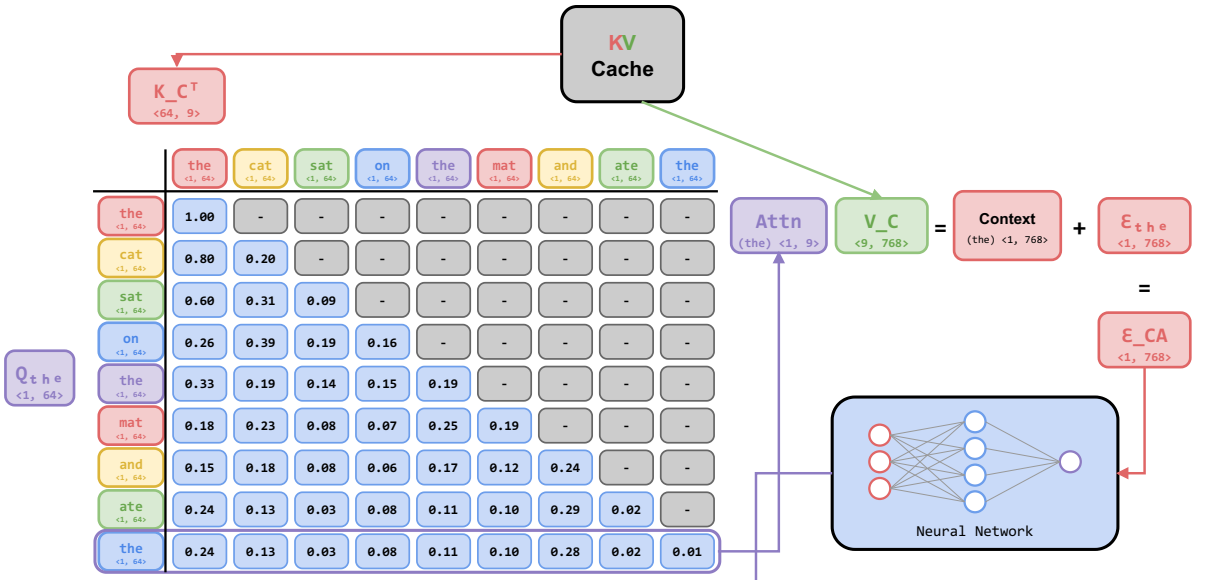
LLM inference can be broken down into two separate steps: **Prefill** and **Decode**

Decode: writing phase where the model **generates the output response** one token at a time, sequentially, making use of the **KV cache** that was **prefilled**

Text Generation with a KV Cache (single head)



Auto-Regressive Text Generation



	0.0304	0.0246	0.0204	0.0184	0.0161	0.0131	0.0117	0.0113	0.0105	0.0101
Token ID	8234	2057	1334	5935	3797	6174	8509	938	717	3290
Token	" fruit"	" food"	" rest"	" egg"	" cat"	" meat"	" bread"	" last"	" first"	" dog"

softmax(Z): the ten most likely next tokens